

Stanley Chow

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EDUCATION & HONORS

The University of Hong Kong

Hong Kong

Bachelor of Science | Major in Mathematics and Computer Science

Jun 2028

GPA: 3.98/4.30 | **Honors:** Dean's List (2025); HKU Undergraduate Entrance Scholarship; Lee Shau Kee Scholarship

Selected Coursework: Machine Learning, Data Structures and Algorithms, Probability, Optimization, Numerical Analysis

PROJECT EXPERIENCE

H&M Two-Stage Personalized Recommendation System

Jul 2026 – Aug 2026

Python, PyTorch, LightGBM, DuckDB | [\[GitHub\]](#)

- Built a two-stage recommender on 31.8M transactions, selecting $K=500$ candidates per customer through nine retrieval channels, then ranking them with LightGBM LambdaRank to generate Top-12 product recommendations.
- Traced ranker collapse to mismatched training negatives and inference candidates. Replacing source-balanced negatives with uniform samples from historical $K=500$ pools raised development MAP@12 from 0.002167 to 0.033679.
- Validated on four rolling cutoffs and an untouched future week (MAP@12 0.034479, 3.94% below development). Scored 1.37M customers in 138 shards, keeping Top-12 results and clearing intermediates with 18.5 GiB peak memory.

WorldQuant International Quant Championship (IQC)

Apr 2026 – Jul 2026

Stage 1 Top 20% Globally; Stage 2 Top 20 in Hong Kong

- Developed and backtested 40+ alphas across momentum, value, volatility and sentiment on WorldQuant BRAIN; submitted 13 high-quality alphas based on platform criteria, backtest performance and inter-alpha correlation.

RESEARCH EXPERIENCE

HKU Summer Research Fellow | Faculty of Science, The University of Hong Kong

May 2026 – Present

Parabolic Inverse Problems and POD Reduced-Order Modeling | *Manuscripts in preparation* | [\[Poster\]](#)

- Built a Python framework for full-order heat-source inversion from noisy observations and rank-20 POD reduction of up to 16,384 state variables; automated Tikhonov-regularized inversion, parameter sweeps and error evaluation.
- Analyzed how regularization, spectral truncation and observation scale affect recovery of source location and coarse shape despite large raw L^2 error. Distinguished raw source, heat-visible field and Hausdorff boundary errors.
- Weighted POD training toward underrepresented source directions and evaluated 672 mesh, parameter and repeated-trial cases under distribution shift. On the largest grid, POD cut query time from 376.67 ms to 107.89 ms (3.49 $\times$).

ADDITIONAL PROJECTS

DocuQuest Agent

Feb 2026 – Mar 2026

C++17, Information Retrieval, CMake | [\[GitHub\]](#)

- Built a C++17 document retrieval engine using a tokenizer, custom BST multimap and inverted index. Matched all terms within each expanded query group, then merged results across groups to select context for answer generation.
- Integrated LLM query expansion and grounded answer generation into the retrieval pipeline. Isolated external model calls behind a client interface, enabling reproducible offline tests of retrieval logic without live model calls.

Flight Price Prediction and Pricing Dynamics

Mar 2026 – Jun 2026

Python, XGBoost, scikit-learn, pandas | [\[GitHub\]](#)

- Developed an XGBoost log-price model and engineered route, airline, cabin-class and booking-time features for a team project on approximately 300K Indian flight records; the pricing-analysis model achieved test R^2 of 0.9688.
- Used partial dependence to compare ticket prices across routes, airlines and cabin classes. Predicted prices were 156.6% higher in economy and 23.5% in business for tickets bought within 3 days of departure versus 20–30 days ahead.

TECHNICAL SKILLS

Programming & Data: Python, C++, SQL, MATLAB; pandas, NumPy, SciPy | **Tools:** Git/GitHub, LaTeX

ML: PyTorch, LightGBM, XGBoost, scikit-learn; regression, classification, feature engineering, model selection

Statistical & Experimental Methods: Hypothesis testing, A/B testing, experimental design, model evaluation

SELECTED HONORS

S.-T. Yau High School Science Award | Semifinalist

May 2023 – Nov 2023